

Spectroscopy as a Probe of Spacetime Foam

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Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, Strehl) constraints. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a spectral shape-distortion bound on the Ng–Perlman foam exponent α . Applied to the MeV emission lines of GRB 221009A in the dynamical-foam limit ($\tau_{\text{foam}} \ll \tau_c$) we obtain $\alpha \gtrsim 0.63$. For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$) with a low-frequency pedestal; a fit to the COBE/FIRAS monopole spectrum reaches $\alpha \gtrsim 0.52$ (95% CL) under the physical Planck-scale pedestal kernel, and next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) are forecast to reach only $\alpha \gtrsim 0.55$, the bound saturating logarithmically as the foam deficit is nearly degenerate with a temperature shift. None of these bounds exclude the holographic value $\alpha = 2/3$, which would require a confirmed narrow spectral line at $E_{\text{obs}} \gtrsim 720$ MeV in a $z \sim 0.15$ GRB.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_P = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler’s original “spacetime foam” picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α -foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha D^{2(1-\alpha)} \ell_P^{2\alpha}, \quad (1)$$

where D is the line-of-sight comoving distance to the source — the appropriate measure rather than the luminosity distance, since the phase accumulation is a property of spacetime itself rather than the photon path [4] — α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_P$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_P}$); holographic foam ($\alpha = 2/3$) grows more slowly, motivated by the heuristic that the information in a spatial region scales with its bounding area rather than its volume — an analogy to the holographic entropy bound, which is rigorous for thermal (black-hole, de Sitter) horizons but is not derived for the random propagation volumes considered here; and the weak-foam limit ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_P$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Light propagating through spacetime foam accumulates a phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_P^\alpha/\lambda$ across the entrance aperture, degrading

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PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6 \mu\text{m}$, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$, has been examined in the imaging channel by Steinbring [11] and Steinbring & Ng [12]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [11] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1^\circ$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of ~ 10 MeV. Steinbring & Ng [12] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ of the observed field and establishes the Gaussian closed form. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem, establishing the split of the observed spectrum into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and a broad pedestal, and connects them to the imaging Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions.

II. SPACETIME FOAM MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t)), \quad (2)$$

where $E_{\text{src}}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for the thermal and incoherent non-thermal sources considered here, including GRB high-energy emission and the CMB) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of E_{src} . Here t is the observer’s time; the deterministic light-travel delay (and cosmological time dilation) is absorbed into the argument of E_{src} , so that $\delta t(t)$ is the zero-mean foam fluctuation about the mean arrival time. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D :

$$\delta t(t) = \frac{1}{c} \Delta\ell_{\text{tot}}(t). \quad (3)$$

The vacuum c in Eq. (3) reflects the standard Ng–Perlman treatment of foam as a small perturbation atop a flat (or FLRW) background, with photons traveling at c along stochastically perturbed geodesics — equivalently, propagation through the effective refractive index $n(\tau, r)$ of Section II B, with $\delta t = \frac{1}{c} \int (n - 1) dr$. In the geometric-optics limit the path-length shift $\Delta\ell_{\text{tot}}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2 . This achromaticity distinguishes foam from other astrophysical screens — such as plasma dispersion, interstellar scintillation, and atmospheric turbulence — all of which imprint frequency-dependent signatures. For broadband sources such as the CMB, the foam distortion is likewise distinct from standard spectral distortions (μ, y) that have different frequency dependences.

B. Metric Origin: Foam as a Time-Varying Refractive Index

To ground the phenomenological jitter of Eq. (3) in the foam metric — and to show that, *within the eikonal reduction adopted here*, its achromaticity is a geometric consequence rather than a separate assumption (its validity is examined at the end of this subsection) — we treat the foam as a stochastic, time-dependent perturbation of a radial Friedmann–Lemaître–Robertson–Walker background. For a radial ray in conformal time τ and comoving radius r ,

$$ds^2 = a^2(\tau) [-(1 + h_{00}) d\tau^2 + (1 + h_{11}) dr^2], \quad (4)$$

with $h_{ab}(\tau, r)$ zero-mean, statistically stationary metric perturbations encoding the Planck-scale foam. The null condition $ds^2 = 0$ gives, to linear order, $dr/d\tau = \sqrt{(1 + h_{00})/(1 + h_{11})} \simeq 1 + \frac{1}{2}(h_{00} - h_{11})$, so the ray propagates with an effective refractive index

$$n(\tau, r) = \left(\frac{dr}{d\tau}\right)^{-1} \simeq 1 - \frac{1}{2}(h_{00} - h_{11}). \quad (5)$$

Because $n - 1$ is geometric — a property of the metric, not of the photon energy — it is frequency-independent, unlike a plasma ($n - 1 \propto \lambda^2$); this is the achromaticity asserted below Eq. (3). The linearization is *per cell* ($|h_{ab}| \ll 1$); the jitter accumulated over the $\sim D/\ell_P$ cells of Eq. (1) can nonetheless be large ($\sigma_\phi^2 \gtrsim 1$; Sec. VB 1). As h_{ab} is zero-mean, $\langle n \rangle = 1$ and the stray $n < 1$ excursions average out: only the second moment $\langle (n - 1)^2 \rangle$ enters the observables. The arrival-time jitter is then the excess optical path,

$$\delta t = \frac{1}{c} \int (n - 1) dr = -\frac{1}{2c} \int (h_{00} - h_{11}) dr, \quad (6)$$

identifying $\Delta \ell_{\text{tot}} = -\frac{1}{2} \int (h_{00} - h_{11}) dr$ in Eq. (3). The integral runs over comoving radius, so $\Delta \ell_{\text{tot}}$ is a comoving path-length fluctuation with variance $\sigma_\ell^2(D_C) = \langle \Delta \ell_{\text{tot}}^2 \rangle$ supplied by the Ng–Perlman relation Eq. (1). Although the path and time jitter are achromatic, the phase variance $\sigma_\phi = (2\pi/\lambda_{\text{obs}}) \sigma_\ell$ that governs every observable inherits the ordinary $1/\lambda_{\text{obs}}$ chromaticity (phase = observed wavenumber $\times$ achromatic path), the scaling $\sigma_\phi \propto 1/\lambda_{\text{obs}}$ used throughout Sec. V; achromatic delay and chromatic phase are thus consistent. What the metric calculation fixes is the *kinematic* map from perturbation to observable; the amplitude and the temporal structure of the correlations (Secs. II C–II D) are modeled.

The frequency shift is fixed by the same h_{ab} . With the observed field $E_{\text{obs}}(t) = E_{\text{src}}(t - \delta t(t))$ of Eq. (2), a monochromatic carrier acquires the instantaneous frequency

$$\frac{\delta \omega}{\omega_0} = -\frac{d\delta t}{dt}, \quad (7)$$

which is exact for the time-warped carrier of Eq. (2): the phase of $E_{\text{src}}(t - \delta t(t))$ advances as $-\omega_0[t - \delta t(t)]$, giving an instantaneous frequency $\omega_0(1 - d\delta t/dt)$, i.e. Eq. (7). Equivalently, it is the leading-order perturbation of the geodesic frequency $\omega = -k_\mu u^\mu$, the homogeneous background contributing the endpoint-determined redshift $\omega \propto a^{-1}$ and the time-dependent perturbation the shift Eq. (7). Phase (through δt) and frequency (through $\delta \omega$) are therefore two projections of the single field $h_{ab}(t)$: a *static* screen ($d\delta t/dt = 0$) shifts no frequency, whereas a time-varying one does, with $\langle \delta \omega \rangle = 0$ and a variance that becomes the line broadening. This is why the bounds rest on the dynamical regime (Section II C).

Equations (5)–(7) place the problem in the standard setting of wave propagation through a random, time-varying refractive medium [13–15]: the phase variance σ_ϕ^2 introduced below is the eikonal phase variance of n , and the temporal autocorrelation $C_\alpha(\tau)$ and differential-jitter variance $\sigma_\Delta^2(\tau)$ of the next two subsections are its mutual-coherence and phase-structure functions. We adopt this eikonal phase-screen reduction throughout. It is the geometric-optics limit of the wave equation in $n(\tau, r)$; with $\tau_{\text{foam}} \sim \ell_P/c$ and foam structure on scales $\ell_P \ll \lambda$, the temporal and transverse variation is strongly non-adiabatic, and a complete wave treatment would add diffractive and inelastic effects beyond the linear phase screen, which we do not model here. In particular, the achromaticity established above holds within this reduction; whether it survives a full sub-wavelength treatment ($\ell_P \ll \lambda$) is beyond our scope.

C. Temporal Decorrelation of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we describe how those fluctuations are correlated across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — determines how the arrival-time jitter $\delta t(t)$ along a single line of sight is correlated across the lag τ . Quantum foam models posit that each Planck-scale cell along the line of sight fluctuates on a characteristic timescale τ_{foam} , so that the jitter at times t and $t + \tau$ is correlated for $\tau \ll \tau_{\text{foam}}$ and essentially independent for $\tau \gg \tau_{\text{foam}}$. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the integrated path-length fluctuation $\Delta \ell_{\text{tot}}(t)$, i.e. $C_\alpha(\tau; D) \equiv \langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle / \sigma_\ell^2(D)$, which inherits the temporal structure of the per-cell foam dynamics over the $\sim D/\ell_P$ cells crossed by the line of sight. Because C_α is defined on the *integrated* path, its temporal shape inherits a residual dependence on D from the way per-cell foam dynamics combine along the line of sight; the explicit D -argument records this dependence. Equivalently,

$$\langle \Delta \ell_{\text{tot}}(t) \Delta \ell_{\text{tot}}(t + \tau) \rangle = \sigma_\ell^2(D) C_\alpha(\tau; D). \quad (8)$$

The timescale τ_{foam} is an additional free parameter of the foam model, not fixed by the α -model variance relation alone. In the standard phenomenological framework [6, 16], each of the $\sim D/\ell_P$ foam cells along the line of sight independently adopts a new configuration on timescale $\tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}$ s; the integrated path-length jitter $\delta \ell_{\text{tot}}(t)$ — a sum of $\sim D/\ell_P$ independent contributions — likewise decorrelates on timescale $\tau_{\text{foam}} \sim \tau_P$. With $C_\alpha(\tau) \approx 0$ for all nonzero accessible lags, the foam is fully decorrelated on any observable timescale — the most optimistic scenario for detection — and for any astrophysical source considered here ($\tau_c/\tau_{\text{foam}} \gtrsim 10^{22}$ even at 37 MeV) the analysis sits deep within the dynamical regime. Models with long-range *temporal* correlations — the temporal analogue of the long-range *transverse* spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give $C_\alpha(\tau) \approx 1$ over macroscopic timescales and thereby drive $\sigma_\Delta^2(\tau) \rightarrow 0$, suppressing the foam signal in both channels analyzed below. The bounds derived here therefore all rest on the dynamical-foam assumption $\tau_{\text{foam}} \ll \tau_c$. We carry $C_\alpha(\tau)$ explicitly only because it sets the pedestal width $\sim 1/\tau_{\text{foam}}$ (Section IV) and the contrast with the quasi-static limit; for the bounds themselves $C_\alpha \rightarrow 0$, an approximation reinforced by transverse decorrelation across the beam (any two rays with $D \delta \theta > \ell_P$ are independent), so the fully-decorrelated single-ray treatment is conservative.

D. Differential Jitter and Its Distribution

With the temporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the differential jitter between photons arriving at times t and $t + \tau$:

$$\Delta(\tau) \equiv \delta t(t + \tau) - \delta t(t). \quad (9)$$

Expanding the square and applying Eq. (8):

$$\begin{aligned} \sigma_\Delta^2(\tau) &= \frac{1}{c^2} \langle [\Delta \ell_{\text{tot}}(t + \tau) - \Delta \ell_{\text{tot}}(t)]^2 \rangle \\ &= \frac{1}{c^2} [\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2 C_\alpha(\tau)] \\ &= \frac{2\sigma_\ell^2(D)}{c^2} [1 - C_\alpha(\tau; D)]. \end{aligned} \quad (10)$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}(0, \sigma_\Delta^2(\tau)), \quad (11)$$

where σ_Δ^2 is taken from Eq. (10); for any source at $D \gtrsim 1$ kpc, the cell count $N_{\text{cells}} \sim D/\ell_P \gtrsim 10^{54}$ amply satisfies the large- N requirement and higher-cumulant corrections are negligible. For $\alpha = 1/2$ (random walk), per-cell path-length variances are equal (ℓ_P^2 each) and the CLT applies in its standard form. For $\alpha \neq 1/2$, the per-cell contributions scale differently with D and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; Gaussianity is adopted as the minimal assumption consistent with the α -model.

The characteristic function of $\Delta(\tau)$ at carrier frequency ω_0 defines a foam suppression factor applicable to any source profile. Since $\Delta(\tau)$ is Gaussian,

$$\langle e^{i\omega_0 \Delta(\tau)} \rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2} \equiv \mathcal{F}_{\text{foam}}(\tau), \quad (12)$$

with $\mathcal{F}_{\text{foam}}(0) = 1$. This foam suppression factor is the dominant term of the transfer function from intrinsic to observed coherence, the higher-cumulant corrections being negligible; the spectroscopic observables below — line broadening, peak suppression, and the CMB core/pedestal split — all follow from it.

From the foam path-length variance $\sigma_\ell^2(D_C)$ and carrier frequency ω_0 , we define $\sigma_\phi^2 \equiv \frac{1}{2}\omega_0^2 \cdot (2\sigma_\ell^2/c^2) = (\omega_0\sigma_\ell/c)^2$ as the asymptotic phase variance — the value $\frac{1}{2}\omega_0^2\sigma_\Delta^2(\tau)$ approaches as $C_\alpha(\tau; D) \rightarrow 0$. In the dynamical-foam limit σ_ϕ^2 is set by the Ng–Perlman path-length variance (1) accumulated along the line-of-sight comoving distance D_C to the source [4]. Substituting $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ from Eq. (10), Eq. (12) takes the compact form

$$\mathcal{F}_{\text{foam}}(\tau) = e^{-\sigma_\phi^2[1 - C_\alpha(\tau; D)]}. \quad (13)$$

At $\tau \rightarrow \infty$: $\mathcal{F}_{\text{foam}}(\infty)$ equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \rightarrow 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$). Because the foam phase is $\phi = \omega_0 \delta t$ — the *observed* carrier $\omega_0 = 2\pi c/\lambda_{\text{obs}}$ times the *observed* arrival-time delay δt — the suppression $e^{-\sigma_\phi^2}$ is manifestly observer-frame and carries no separate cosmological reweighting; the comoving distance enters only through the Ng–Perlman variance $\sigma_\ell^2(D_C)$, so

$$\sigma_\phi^2 = \left(\frac{2\pi}{\lambda_{\text{obs}}}\right)^2 \sigma_\ell^2(D_C), \quad \sigma_\phi = \frac{2\pi}{\lambda_{\text{obs}}} \sigma_\ell, \quad (14)$$

the *observed* wavenumber $2\pi/\lambda_{\text{obs}}$ times the achromatic path rms $\sigma_\ell(D_C)$ ($A_\alpha = 1$). The three foam variances are one quantity in different dress: the geometric $\sigma_\ell(D_C)$, the differential arrival-time jitter $\sigma_\Delta = \sqrt{2}\sigma_\ell/c$ (Eq. (10)), and the phase $\sigma_\phi = \omega_0\sigma_\ell/c$. We adopt no $(1+z')^2$ path reweighting: any cosmological structure is already carried by $\sigma_\ell^2(D_C)$ through the comoving distance.

III. FIRST-ORDER COHERENCE

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g_{\text{obs}}^{(1)}(\tau) = \frac{\langle E_{\text{obs}}^*(t) E_{\text{obs}}(t + \tau) \rangle}{\langle |E_{\text{obs}}(t)|^2 \rangle}. \quad (15)$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t + \tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of E_{src} gives

$$\langle E_{\text{src}}^*(u) E_{\text{src}}(v) \rangle = \langle |E_{\text{src}}|^2 \rangle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)). \quad (16)$$

Since δt is independent of E_{src} , and since $\langle |E_{\text{obs}}(t)|^2 \rangle = \langle |E_{\text{src}}(t - \delta t)|^2 \rangle = \langle |E_{\text{src}}|^2 \rangle$ by stationarity (so the normalization in Eq. (15) cancels), averaging over the jitter yields

$$g_{\text{obs}}^{(1)}(\tau) = \langle g_{\text{src}}^{(1)}(\tau - \Delta(\tau)) \rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau)) g_{\text{src}}^{(1)}(\tau - \Delta) d\Delta. \quad (17)$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [17], and it serves as the representative case admitting the simplest closed form. A Gaussian line of 1σ width σ_ν centered on ν_0 corresponds via the Wiener–Khinchin theorem to the coherence function

$$g_{\text{src}}^{(1)}(\tau) = e^{-i\omega_0\tau} e^{-\tau^2/2\tau_G^2}, \quad (18)$$

where $\omega_0 \equiv 2\pi\nu_0$ and $\tau_G \equiv 1/(2\pi\sigma_\nu)$ is the Gaussian coherence decay time. The carrier factor $e^{-i\omega_0\tau}$ encodes the line’s position in the spectrum and must be retained: dropping it would substitute a baseband Gaussian centered at $\nu = 0$ for the physical line and would lose the dominant foam effect (line-peak suppression). Substituting into the jitter-convolution integral (17) and completing the square gives the exact closed form

$$\begin{aligned} g_{\text{obs}}^{(1)}(\tau) &= \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}} \exp\left[-i\omega_0\tau \frac{\tau_G^2}{\tau_G^2 + \sigma_\Delta^2(\tau)}\right] \\ &\times \exp\left[-\frac{\tau^2 + \omega_0^2 \tau_G^2 \sigma_\Delta^2(\tau)}{2[\tau_G^2 + \sigma_\Delta^2(\tau)]}\right]. \end{aligned} \quad (19)$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight redshift of the carrier, $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / [\tau_G^2 + \sigma_\Delta^2(\tau)]$; (ii) a wider Gaussian decay of $g_{\text{obs}}^{(1)}(\tau)$ in lag (decay scale $\sqrt{\tau_G^2 + \sigma_\Delta^2(\tau)}$ vs. τ_G); and (iii) a global amplitude suppression by the factor $\exp[-\omega_0^2 \tau_G^2 \sigma_\Delta^2 / \{2(\tau_G^2 + \sigma_\Delta^2)\}]$. Normalization $g_{\text{obs}}^{(1)}(0) = 1$ holds because $\sigma_\Delta(0) = 0$. The narrowband limit $\tau_G \rightarrow \infty$ of Eq. (19) reduces to a pure phase-modulated oscillation, which is the single-mode building block from which the broadband CMB spectrum of Section V B 2 is constructed by incoherent superposition.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (17) with the closed form Eq. (19) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\text{obs}}(\nu) = \mathcal{F}\{g_{\text{obs}}^{(1)}(\tau)\}(\nu). \quad (20)$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on light after emission, while $\delta t(t)$ statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time τ_{foam} (the lag scale beyond which $C_\alpha(\tau; D) \rightarrow 0$), so that the time-averaged spectrum coincides with its ensemble average: $T_{\text{int}} \gg \tau_{\text{foam}}$.

In the dynamical limit $C_\alpha(\tau; D) \rightarrow 0$ for $|\tau| \gtrsim \tau_{\text{foam}}$, $\mathcal{F}_{\text{foam}}(\tau)$ acquires a long-lag plateau $e^{-\sigma_\phi^2}$ outside a narrow window $|\tau| \lesssim \tau_{\text{foam}}$. Fourier-transforming via Eq. (20), the plateau produces a coherent core $e^{-\sigma_\phi^2} S_{\text{src}}(\nu)$, while the short-lag window redistributes the complementary fraction $1 - e^{-\sigma_\phi^2}$ of the source flux into a broad pedestal. This core-plus-pedestal split holds for any source, with σ_ϕ^2 evaluated frequency by frequency ($\sigma_\phi^2 \rightarrow \sigma_\phi^2(\nu)$) for a broadband source; its broadband form for the CMB, including the pedestal shape, is worked out for a specific model of the temporal autocorrelation C_α in Section V B 2.

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (19) with σ_Δ^2 held at its asymptotic value $2\sigma_\ell^2/c^2$ gives the explicit Gaussian core

$$S_{\text{obs}}^{\text{core}}(\nu) = \frac{\tau_G}{\sqrt{\tau_G^2 + \sigma_\Delta^2}} \exp\left[-\frac{\sigma_\phi^2 \tau_G^2}{\tau_G^2 + \sigma_\Delta^2}\right] \cdot \frac{1}{\sqrt{2\pi} \sigma_{\text{obs}}} \exp\left[-\frac{(\nu - \nu_0^{\text{eff}})^2}{2\sigma_{\text{obs}}^2}\right], \quad (21)$$

with $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / (\tau_G^2 + \sigma_\Delta^2)$ and $\sigma_{\text{obs}} = 1/[2\pi \sqrt{\tau_G^2 + \sigma_\Delta^2}] < \sigma_\nu$. The Gaussian's carrier shift, line narrowing, and peak suppression are the frequency-domain images of items (i)–(iii) of Section III B.

The pedestal in either decomposition is sourced by the short-lag window $|\tau| \lesssim \tau_{\text{foam}}$ and so spreads over a frequency range $\sim 1/\tau_{\text{foam}}$. When $\tau_{\text{foam}} \ll \tau_G, \sigma_\Delta$ the pedestal is correspondingly broad and shallow, and the spectrum near the line peak is dominated by the coherent core.

B. Connection to the Imaging Strehl Ratio

The first-order coherence function $g_{\text{obs}}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (13). The Strehl ratio of an imaging telescope is the *spatial* counterpart: phase variations across the aperture plane are governed by an aperture-plane coherence function $C_\alpha^{\text{sp}}(B; D)$, constructed by direct analogy to the temporal $C_\alpha(\tau; D)$ of Section II C under the same foam-cell statistics.

The structural equivalence is not merely formal: in both cases the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable. Temporally, the variable is the differential jitter $\Delta(\tau) = \delta t(t + \tau) - \delta t(t)$ with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (10)). Spatially, the variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x} + \mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B , with variance

$$\sigma_{\Delta\phi}^2(B) = 2\sigma_\phi^2[1 - C_\alpha^{\text{sp}}(B; D)]. \quad (22)$$

The focal-plane field of a telescope is the Fraunhofer Fourier transform of the aperture-plane field, giving the diffraction-limited PSF when the aperture phase is uniform. Foam imprints a Gaussian random phase $\phi(\mathbf{x})$ on the aperture with variance σ_ϕ^2 and two-point correlation $C_\alpha^{\text{sp}}(B; D)$, blurring this PSF. In the resolved limit $C_\alpha^{\text{sp}}(D_{\text{tel}}; D) \rightarrow 0$ — where aperture phases at distinct points decorrelate — the on-axis peak is attenuated by the Strehl ratio

$$S \simeq e^{-\sigma_\phi^2} \quad (23)$$

(extended-Maréchal [4]), the surrounding Airy fringes are washed out by the same factor, and the complementary $1 - e^{-\sigma_\phi^2}$ of the source flux scatters into an incoherent halo.

Imaging and spectroscopic channels each probe σ_ϕ^2 through a Fourier transform — aperture position $\leftrightarrow$ focal-plane angle in imaging (Fraunhofer), lag $\leftrightarrow$ frequency in spectroscopy (Wiener-Khinchin) — with the rapid chromatic tightening $\sigma_\phi \propto 1/\lambda$ exploited in both Ng & Perlman’s [4] Table 1 and the analyses of Section V A. The shared σ_ϕ^2 that suppresses the coherent diffraction-limited PSF in imaging also suppresses the coherent line core in spectroscopy, placing the published imaging constraints [4, 7–9] on a common footing with the present framework. A quantitative comparison across channels is, however, model-dependent: it requires relating the spatial (C_α^{sp}) and temporal (C_α) foam correlation functions under a common foam-cell model, so the two constraints coincide only to the precision of those shared model details. The duality is not perfect at the observational interface: in the imaging channel the reference truth—the telescope aperture—is known a priori, whereas in the spectroscopic channel the corresponding truth—the intrinsic source spectrum—is itself uncertain, so the foam contribution is harder to isolate.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are distorted according to the Gaussian-source result Eq. (21), with a carrier shift, line narrowing, and peak suppression that scale chromatically as $\sigma_\phi \propto 1/\lambda$. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton- y distortions. In this section we consider these two possibilities.

A. GRB 221009A

For spectroscopy of a single source without an absolute flux reference, the smooth suppression factor $e^{-\sigma_\phi^2(\nu)}$ attenuates line and continuum equally at any given frequency, preserving the equivalent width: uniform dimming is undetectable from the spectrum alone. The carrier-included closed form Eq. (19) supplies the needed shape-distortion criterion: foam broadens the observed line width by a calculable energy-dependent amount (derived below as Eq. (24)), and multi-epoch line-width measurements across a wide energy range give additional leverage, as we now show for GRB 221009A. We first establish the multi-epoch line dataset, decompose the observed fractional width into intrinsic and foam contributions, fit the two-parameter model to obtain an upper limit on the foam amplitude, and convert it into a lower bound on α .

GRB 221009A (“the BOAT,” Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\text{iso}} \sim 10^{55}$ erg at redshift $z = 0.151$, a once-in- $\sim 10^4$ -year event [18]. Two independent spectral analyses have reported a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron-positron pair annihilation [17, 19], at observed energies spanning 6–37 MeV. This pair-annihilation identification is one proposed interpretation and is not yet definitive; the foam analysis depends only on Doppler-invariance of the line’s fractional width, not on the specific emission mechanism. We exploit the *multi-epoch* constancy of the fractional line width to obtain a stringent constraint; the seven sub-intervals with well-constrained widths span the 9–37 MeV lever arm (a factor of ≈ 4 in energy), within the broader 6–37 MeV detection range. Adopting flat Λ CDM ($\Omega_M = 0.27$, $\Omega_\Lambda = 0.73$, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the comoving distance is $D_C(z = 0.151) \approx 0.63 \text{ Gpc} \approx 1.93 \times 10^{25} \text{ m}$ and $\ln(D_C/\ell_P) \approx 138.3$.

The analysis below operates in the dynamical-foam regime $\tau_{\text{foam}} \ll \tau_c$ adopted throughout Section II; in the opposite quasi-static limit the framework yields no foam imprint on the line spectrum and the bound does not apply (Section VI).

Multi-epoch line data. Zhang et al. [17] perform a joint analysis of GECAM-C and Fermi/GBM data covering the full prompt emission, including the initial bright phase that saturated GBM alone. They detect a Gaussian emission line with centroid evolving as $E_{\text{line}} \propto t^{-1}$ from $\approx 37 \text{ MeV}$ down to $\approx 6 \text{ MeV}$ over $T_{\text{ref}} + 246\text{--}360 \text{ s}$. In the seven sub-intervals

where the line is sufficiently bright to constrain its width, the measured fractional width $r_{\text{obs}} \equiv \sigma_{\text{obs}}/E_{\text{line}} = 0.10 \pm 0.01$ is consistent with a single constant across the full dynamic range ($\chi^2/\text{dof} = 4/7$; Zhang et al. §4.4 and their Figure 7c). This constancy is the predicted behavior of a Doppler-boosted line. In the comoving frame of the emitting plasma the line has fractional width $r' = \sigma'/E'$, fixed by the comoving emission physics (pair-plasma temperature, velocity dispersion). When the outflow has bulk Doppler factor $\mathcal{D}$ — epoch-dependent as the jet decelerates — both the line centroid and its width are multiplied by the same factor: $E = \mathcal{D}E'$, $\sigma_{\text{int}} = \mathcal{D}\sigma'$, so $r_{\text{int}} \equiv \sigma_{\text{int}}/E = r'$. The Doppler factor cancels in the ratio; r_{int} is a Lorentz-invariant property of the line, independent of epoch. The observed $E_{\text{line}} \propto t^{-1}$ centroid evolution over a factor of ≈ 6 in energy traces the bulk-Doppler deceleration of the jet, and the simultaneous constancy of r_{obs} confirms the prediction. The line is independently detected by Ravasio et al. [19] in Fermi/GBM data alone, with $\sigma_{\text{gauss}} = 1.31^{+0.31}_{-0.30}$ MeV at $E_{\text{line}} = 12.56^{+0.30}_{-0.31}$ MeV in $T + 280\text{--}300$ s ($\sigma/E \approx 10.4\%$, consistent with Zhang's $r_{\text{obs}} = 0.10 \pm 0.01$ at 0.4σ in the time interval that overlaps Zhang's bright-line window). Ravasio also reports the line in three later intervals ($T + 300\text{--}360$ s) where σ/E rises to $\approx 15\text{--}20\%$, so the constancy of r_{obs} across the full 9–37 MeV lever arm rests on the GECAM-C+Fermi/GBM joint analysis [17], with Ravasio providing an instrument-independent confirmation of the line at the highest-signal-to-noise epoch.

Width decomposition. Because r_{int} is the same in every epoch, the multi-epoch measurements constrain a single common r_{int} alongside the foam amplitude $r_{\text{foam},0}$, giving the lever arm its statistical power. The Gaussian-source spectrum Eq. (21) gives the observed centroid and absolute width $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / (\tau_G^2 + \sigma_\Delta^2)$ and $\sigma_{\text{obs}} = 1/[2\pi\sqrt{\tau_G^2 + \sigma_\Delta^2}]$. Although σ_{obs} is slightly *narrower* than the intrinsic σ_ν by a factor $\tau_G/\sqrt{\tau_G^2 + \sigma_\Delta^2}$, the centroid is redshifted by the *square* of that factor, so the fractional width that Zhang et al. report as σ/E in each epoch *increases*:

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 [1 + \sigma_\Delta^2 / \tau_G^2(E)] = r_{\text{int}}^2 + r_{\text{foam}}^2(E), \quad (24)$$

with foam excess $r_{\text{foam}}^2(E) \equiv r_{\text{int}}^2 \sigma_\Delta^2 / \tau_G^2(E)$. The relation is multiplicative in $(1 + \sigma_\Delta^2 / \tau_G^2)$, not additive in independent foam/intrinsic widths: the foam acts on the source through the jitter-convolution Eq. (17), a *multiplication* of the spectrum by a Gaussian factor rather than a convolution with it (Section III B).

In the dynamical-foam limit ($\tau_{\text{foam}} \ll \tau_c$), $\sigma_\Delta^2(\tau)$ saturates to $2\sigma_\ell^2/c^2$ on lags $|\tau| \gtrsim \tau_{\text{foam}}$. The broadening term in Eq. (24) is set by this variance at the source coherence lag $\tau \sim \tau_G$, which satisfies $\tau_G \sim \tau_c \gg \tau_{\text{foam}}$ in this regime, so the saturated value is the operative one. The dominant foam effect on the spectrum is then carrier-phase suppression of the line peak by $e^{-\sigma_\phi^2(E)}$, together with line-width broadening captured by item (ii) of Section III B; both follow from Eq. (19). The observed arrival-time jitter is $\sigma_\Delta = \sqrt{2}\sigma_\ell/c$ (Eq. (10)); with $\tau_G(E) = \lambda_{\text{obs}}(E)/(2\pi c r_{\text{int}})$ the foam-to-intrinsic timescale ratio takes the compact form

$$\frac{\sigma_\Delta}{\tau_G(E)} = \sqrt{2} r_{\text{int}} \sigma_\phi(E), \quad (25)$$

the product of the intrinsic fractional width r_{int} (a source property) and the phase variance σ_ϕ (Eq. (14)). Substituting into Eq. (24) gives $r_{\text{obs}}^2(E) = r_{\text{int}}^2 [1 + 2r_{\text{int}}^2 \sigma_\phi^2(E)]$, so the foam excess is

$$r_{\text{foam}}(E) = \sqrt{2} r_{\text{int}}^2 \sigma_\phi(E) = r_{\text{int}}^2 \cdot \frac{2\sqrt{2}\pi\sigma_\ell}{\lambda_{\text{obs}}(E)} \propto E. \quad (26)$$

The r_{int}^2 prefactor reflects two geometric factors: one converts the absolute foam-induced shift ($\propto \sigma_\phi \nu_0$) into a fractional width; the other enters through $\tau_G \propto 1/r_{\text{int}}$ in the broadening term σ_Δ^2/τ_G^2 . A narrower intrinsic line (smaller r_{int} , longer τ_G) is therefore more robust to dynamical foam: σ_Δ falls further below the source coherence time.

The fit-ready two-parameter model is

$$r_{\text{obs}}^2(E) = r_{\text{int}}^2 + r_{\text{foam},0}^2 (E/E_0)^2, \quad (27)$$

with $r_{\text{foam},0} \equiv r_{\text{foam}}(E_0)$ the foam contribution at the reference energy $E_0 = 37$ MeV (observed wavelength $\lambda_{\text{obs}}(E_0) = hc/E_0 \approx 3.35 \times 10^{-14}$ m).

Fit and upper limit on $r_{\text{foam},0}$. A χ^2 fit of the seven Zhang et al. measurements ($E_i, r_i \pm \delta r_i$) to Eq. (27) with free parameters ($r_{\text{int}}, r_{\text{foam},0} \geq 0$) yields a best fit at $r_{\text{foam},0} = 0$, $r_{\text{int}} = 0.111$, with $\chi_{\text{min}}^2 = 5.3$ ($\chi^2/\text{dof} = 0.88$ with 6 effective degrees of freedom; since $r_{\text{foam},0} = 0$ at the boundary the fit reduces to one free parameter r_{int}), confirming the constant- r_{obs} hypothesis adopted by Zhang et al. (their §4.4 and Figure 7c). Profiling over r_{int} gives the one-sided 95% CL upper limit $r_{\text{foam},0}^{\text{max}} = 0.27$ at $\Delta\chi^2 = 2.71$. [20] With seven measurements and per-point fractional uncertainties of 30–50%, the $\Delta\chi^2 = 2.71$ coverage relies on the Gaussian- χ^2 approximation.

Constraint on α . Eq. (26) gives $\sigma_\ell = r_{\text{foam},0} \lambda_{\text{obs}} / (2\sqrt{2}\pi r_{\text{int}}^2)$; converting via $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ (Eq. (1) with $A_\alpha = 1$) gives

$$\alpha \gtrsim \frac{\ln(D_C/\lambda_{\text{obs}}) + |\ln r_{\text{foam},0}^{\text{max}}| - |\ln(2\sqrt{2}\pi r_{\text{int}}^2)|}{\ln(D_C/\ell_P)} = \frac{89.3 + 1.3 - 3.3}{138.3} \approx 0.63. \quad (28)$$

Using $\sigma_\Delta = \sqrt{2}\sigma_\ell/c$ and $c\tau_G(E_0) = \hbar c/(r_{\text{int}}E_0) \approx 8.1 \times 10^{-14}$ m for $r_{\text{int}} = 0.066$, the ratio $\sigma_\Delta/\tau_G(E_0) \approx 4.1$ at the bound ($\sigma_\ell \approx 2.3 \times 10^{-13}$ m), placing the bound deep in the strong-foam regime $\sigma_\Delta \gtrsim \tau_G$. The individual line-width measurements carry 30–50% fractional uncertainties; the multi-epoch lever arm is essential for reaching this precision.

The G1 interval ($T+246\text{--}256$ s, $E \approx 37$ MeV) lies at or beyond the upper boundary of the GBM BGO detector range (4–35 MeV), so its line-width measurement carries larger systematic uncertainty than the later intervals. Refitting the remaining six intervals gives a best fit at $r_{\text{foam},0} = 0$, $r_{\text{int}} = 0.109$, with $\chi^2_{\text{min}} = 4.4$ over five effective degrees of freedom, and a 95% CL profile boundary at $r_{\text{foam},0}^{\text{max}} = 0.27$ paired with $r_{\text{int}} = 0.066$ — yielding $\alpha \gtrsim 0.63$, statistically indistinguishable from the all-seven-interval bound. G1’s large fractional uncertainty ($\sim 40\%$) gives it little weight in the χ^2 despite its position at the top of the lever arm.

At the bound ($\sigma_\Delta \gtrsim \tau_G$), Eq. (19) predicts two effects: line-width broadening — which the fit constrains via $r_{\text{foam},0}^{\text{max}} = 0.27$ — and a line-center redshift $\nu_0^{\text{eff}}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ that becomes substantial there ($\nu_0^{\text{eff}}/\nu_0 \approx 0.06$ at $\sigma_\Delta/\tau_G \approx 4.1$). The lever-arm fit constrains only the former, using observed line centers as if intrinsic. Closing the gap requires either (i) an independent constraint on the per-epoch bulk Doppler factor $\mathcal{D}_i$ — from afterglow blastwave modeling, TeV-photon photosphere arguments, or variability-timescale estimates — that pins the pre-foam centroid, or (ii) detection of a second emission line of known rest-frame energy whose distinct $\tau_G(E)$ would produce a different foam-redshift factor in the same epoch and thereby separate $\mathcal{D}_i$ from the foam shift. Neither is available in the present dataset. The bound is therefore weaker than what a joint fit would yield, and at the upper limit it is moreover not internally self-consistent: with $\sigma_\Delta^2/\tau_G^2 \propto E^2$, the foam-induced centroid factor ν_0^{eff}/ν_0 would vary across the data range from ≈ 0.06 at $E = 37$ MeV to ≈ 0.7 at $E = 6$ MeV, producing a strongly energy-dependent depression of the observed centroid that is visibly inconsistent with the smooth $E_{\text{line}} \propto t^{-1}$ Doppler-deceleration trend in the data. A joint center-plus-width fit would quantify this and likely tighten the bound substantially.

If the intrinsic fractional width r_{int} evolves with energy (e.g. due to changing pair-plasma temperature across epochs), this could partially mimic or mask a foam signal. Marginalizing over a free linear-in- E nuisance term $r_{\text{int}}(E) = r_0 + r_1(E/E_0)$ leaves the bound essentially unchanged ($\alpha \gtrsim 0.63$), because the E^2 foam scaling has distinct curvature from a linear intrinsic trend across the lever arm. Numerically, the best-fit linear slope r_1 is consistent with zero and, although the 95% CL upper limit on $r_{\text{foam},0}$ broadens from 0.27 to 0.31 when the nuisance is included, the corresponding α bound shifts by $\Delta\alpha \approx 3 \times 10^{-3}$ — a direct consequence of the logarithmic compression in the Ng–Perlman relation — confirming that the E^2 and linear- E shapes are effectively orthogonal for the seven-point lever-arm spanning 9–37 MeV.

B. CMB

For a broadband thermal source such as the CMB, foam does not introduce a spectral-line shift but instead redistributes power: the coherent-core component is suppressed by $e^{-\sigma_\phi^2(\nu)}$ and the missing flux reappears in a broad pedestal whose shape depends on the short-time behavior of the correlation function $C_\alpha(\tau)$. Because $\sigma_\phi^2 \propto \nu^2$ (Eq. (14)), the foam deficit is concentrated at high frequencies while the pedestal fills in at low frequencies, producing a spectral signature distinct from standard chemical-potential (μ) and Compton- y distortions.

The Far-InfraRed Absolute Spectrophotometer (FIRAS) aboard the Cosmic Background Explorer (COBE) measured the absolute spectrum of the CMB monopole over 60–600 GHz by direct comparison against an internal cryogenic blackbody calibrator — the absolute-photometry feature that distinguishes it from relative-photometry experiments and makes it well suited to constraining spectral *shape* distortions. The calibrated monopole agrees with a pure blackbody to within ~ 50 ppm (a few $\times 10^{-5}$ fractional) across the band, placing the still-standard limits $|\mu| \lesssim 9 \times 10^{-5}$ and $|y| \lesssim 1.5 \times 10^{-5}$ [21]. These limits and the 43-channel calibrated spectrum form the empirical basis for the foam constraint derived in this subsection, which sets up the broadband signal, derives it from first principles, and analyzes the FIRAS sensitivity.

1. Foam phase variance at CMB frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_C = 13.87$ Gpc (Planck 2018 flat- Λ CDM cosmology; $H_0 = 67.36$ km s $^{-1}$ Mpc $^{-1}$, $\Omega_m = 0.315$). The foam phase variance follows Eq. (14) with the observed wavelength; the cumulative comoving foam variance $A_{\text{eff}} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha} = \sigma_\ell^2(D_C)$ used in Section V B 3 (no $(1+z)^2$ path reweighting) enters as

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c} \right)^2 A_{\text{eff}}, \quad (29)$$

with $A_\alpha = 1$.

At $\nu = 150$ GHz this gives

$$\sigma_\phi^2(150 \text{ GHz}) \approx \begin{cases} 7 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 2 \times 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \quad (30)$$

The strong ν^2 dependence concentrates foam effects at high frequency; for $\alpha = 1/2$ the variance reaches unity only near the top of the FIRAS band ($\nu \approx 570$ GHz), marking where the linearised template breaks down (Section V B 3).

2. Foam-modified CMB spectrum

The broadband blackbody admits a direct evaluation: compute $g_{\text{src}}^{(1)}(\tau)$ as the Fourier transform of $B_\nu(T_0)$ via Eq. (20), apply the jitter-convolution Eq. (17), and inverse-transform to recover $S_{\text{obs}}(\nu)$. This route entangles the model-independent per-mode suppression $e^{-\sigma_\phi^2(\nu)}$ with the C_α -dependent redistribution of the suppressed flux. We instead present a derivation that isolates these two contributions: the coherent core, which depends only on the asymptotic phase variance $\sigma_\phi^2(\nu)$ and is therefore obtained in closed form independent of foam temporal dynamics, and the pedestal kernel $\hat{R}(\nu - \nu'; \nu')$, which carries the full dependence on the short- τ behavior of $C_\alpha(\tau)$ and is treated phenomenologically pending a first-principles specification of the foam temporal autocorrelation. A worked example with exponential $C_\alpha(\tau)$ below makes the pedestal kernel concrete and identifies the physically-realized case used in the Fisher analysis of Section V B 3.

Decomposing the blackbody as an incoherent superposition of narrowband modes $S_{\text{src}}(\nu') = B_{\nu'}(T_0)$, each mode is modulated by the *same* achromatic jitter $\delta t(t)$, with per-mode foam suppression

$$\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}, \quad \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D), \quad (31)$$

that approaches the constant $e^{-\sigma_\phi^2(\nu')}$ for all $\tau \neq 0$ in the $C_\alpha \rightarrow 0$ limit. For the CMB line of sight, $\sigma_\ell^2(D)$ here is the comoving cumulative variance of Section V B 1, so that $\sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 A_{\text{eff}}$, the cumulative comoving foam variance used in the fit.

For a single narrowband mode at frequency ν' , the source coherence is $g_{\text{src}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau}$ (the $\tau_G \rightarrow \infty$ limit of Eq. (18)), and the jitter convolution gives

$$g_{\text{obs}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau} \mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-i2\pi\nu'\tau} e^{-\sigma_\phi^2(\nu')[1-C_\alpha(\tau)]}.$$

In the dynamical limit, $\mathcal{F}_{\text{foam}}(\tau; \nu')$ transitions from $\mathcal{F}_{\text{foam}}(0; \nu') = 1$ to the plateau $e^{-\sigma_\phi^2(\nu')}$ within the narrow window $|\tau| \lesssim \tau_{\text{foam}}$. The Fourier transform of the plateau (constant in τ outside that window) is $e^{-\sigma_\phi^2(\nu')} \delta(\nu - \nu')$ — the coherent core at this mode's frequency — while the Fourier transform of the rise from plateau to unity is supported on $|\tau| \lesssim \tau_{\text{foam}}$ and is therefore broad in frequency, of width $\sim 1/\tau_{\text{foam}}$ — the pedestal at this mode. The split applied below to $\mathcal{F}_{\text{foam}}(\tau; \nu')$ generalizes this single-mode decomposition to the full Planck spectrum by linear superposition.

To isolate the coherent-core and pedestal contributions, we split $\mathcal{F}_{\text{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')} + [\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_\phi^2(\nu')}]$: the first term contributes the constant asymptote at all lags (Fourier-transforming to a delta function at $\nu = \nu'$), while the residual is localized at $|\tau| \lesssim \tau_{\text{foam}}$ (Fourier-transforming to a broad pedestal). Taking the intensity-weighted Fourier transform via Eq. (20) yields

$$S_{\text{obs}}(\nu) = e^{-\sigma_\phi^2(\nu)} B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu', \quad (32)$$

where $\hat{R}$ denotes the residual's Fourier transform. The coherent-core term $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$ is exact and model-independent (requiring only achromatic jitter and $C_\alpha \rightarrow 0$); the pedestal kernel $\hat{R}$ depends on the short- τ behavior of $C_\alpha(\tau)$ — the foam coherence timescale τ_{foam} — and is therefore undetermined without specifying C_α . This is the broadband generalization of the core-plus-pedestal decomposition stated in Section IV. Because the per-mode phase modulation $e^{i\omega\delta t(t)}$ is unitary, any physical pedestal kernel $\hat{R}$ derived from a specific $C_\alpha(\tau)$ conserves photon number per mode: $\int [S_{\text{obs}}(\nu) - B_\nu(T_0)] d\nu/(h\nu) = 0$.

To make the pedestal kernel concrete, we evaluate $\hat{R}$ for the minimal one-parameter model $C_\alpha(\tau) = e^{-|\tau|/\tau_{\text{foam}}}$, the autocorrelation of a Markovian metric fluctuation. The framework does not require this specific choice; any $C_\alpha(\tau)$ with characteristic decorrelation time $\tau_{\text{foam}} \ll 1/\nu_{\text{max}}$ (satisfied with enormous margin for Planck-scale foam,

$\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44} \text{ s}$) has a Fourier transform of width $\gtrsim 1/\tau_{\text{foam}} \gg \nu_{\text{max}}$, so the pedestal kernel is constant across the FIRAS band irrespective of its functional form, and the conclusion that the in-band pedestal is negligible follows from the bandwidth ratio alone. Substituting into Eq. (10) gives

$$\sigma_{\Delta}^2(\tau) = \frac{2\sigma_{\ell}^2(D)}{c^2} [1 - e^{-|\tau|/\tau_{\text{foam}}}] \quad (33)$$

The Fisher analysis below operates in the linearised regime $\sigma_{\phi}^2(\nu') \ll 1$. Expanding $\mathcal{F}_{\text{foam}}(\tau; \nu')$ to first order and subtracting the long-lag asymptote $e^{-\sigma_{\phi}^2(\nu')}$ leaves the short- τ residual

$$\mathcal{F}_{\text{foam}}(\tau; \nu') - e^{-\sigma_{\phi}^2(\nu')} \approx \sigma_{\phi}^2(\nu') e^{-|\tau|/\tau_{\text{foam}}}, \quad (34)$$

whose Fourier transform is a Lorentzian centered on the source frequency,

$$\hat{R}(\nu - \nu'; \nu') = \sigma_{\phi}^2(\nu') \frac{2\tau_{\text{foam}}}{1 + [2\pi(\nu - \nu')\tau_{\text{foam}}]^2}, \quad (35)$$

of full width at half maximum $1/(\pi\tau_{\text{foam}})$. For Planck-scale foam $\tau_{\text{foam}} \sim \ell_P/c \approx 5 \times 10^{-44} \text{ s}$ the Lorentzian width is $\sim 6 \times 10^{42} \text{ Hz}$, eleven orders of magnitude beyond the FIRAS upper edge $\nu_{\text{max}} = 6 \times 10^{11} \text{ Hz}$; the fraction of the redistributed flux falling within the FIRAS band is $\sim 10^{-31}$. The physical pedestal therefore deposits no detectable in-band power for any plausible τ_{foam} . The constraint of Section V B 3 consequently adopts the no-pedestal limit as the primary template.

To probe robustness against in-band pedestal contamination beyond the physical-kernel result above, we also evaluate the Fisher matrix with a μ -aligned band-filling redistribution kernel, deliberately shaped $\propto 1/\nu$ to match the Rayleigh–Jeans form of the μ -distortion template (Section V B 3) and thereby maximize the foam- μ Fisher degeneracy — the most effective way an in-band pedestal could disguise foam as a standard μ -distortion:

$$K_{\nu}^{\text{fast}}(\nu) = \begin{cases} \frac{1}{\nu \ln(\nu_{\text{max}}/\nu_{\text{min}})} & \nu_{\text{min}} \leq \nu \leq \nu_{\text{max}}, \\ 0 & \text{otherwise,} \end{cases} \quad (36)$$

with $\nu_{\text{min}} = 60 \text{ GHz}$ and $\nu_{\text{max}} = 600 \text{ GHz}$ set by the FIRAS band edges. $K_{\nu}^{\text{fast}}(\nu)$ is a normalized density in observer frequency ($\int_{\nu_{\text{min}}}^{\nu_{\text{max}}} K_{\nu}^{\text{fast}} d\nu = 1$, units Hz^{-1}) that specifies the fraction of the total foam-removed flux reappearing as pedestal intensity per Hz at observer frequency ν ; the form above corresponds to uniform energy redistribution in $\ln \nu$ across the band. Within the FIRAS band, $K_{\nu}^{\text{fast}}(\nu)$ replaces the physical pedestal contribution to Eq. (32) by a μ -aligned factorised substitution

$$\int B_{\nu'}(T_0) \hat{R}(\nu - \nu'; \nu') d\nu' \longrightarrow \sigma_{\ell}^2(D) K_{\nu}^{\text{fast}}(\nu) P_{\text{foam}}, \quad P_{\text{foam}} \equiv \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left(\frac{2\pi\nu'}{c} \right)^2 B_{\nu'}(T_0) d\nu', \quad (37)$$

with P_{foam} the total foam-removed flux per unit σ_{ℓ}^2 . The substitution forces the pedestal kernel to factorise into a ν -only template (K_{ν}^{fast}) times a ν' -only weighting, a structure no physical $\hat{R}(\nu - \nu'; \nu')$ satisfies (the Lorentzian above carries ν' -dependence through $\sigma_{\phi}^2(\nu')$). The template is energy-conserving but not photon-number-conserving, and serves purely as a μ -aligned probe of the foam- μ kernel-shape degeneracy. Its effect on the bound is evaluated in Section V B 4.

3. Constraint from FIRAS

We use $I_{\nu} \equiv S_{\text{obs}}(\nu)$ to denote the observed CMB intensity including the foam signal. Working in the linearised regime $\sigma_{\phi}^2(\nu) \ll 1$, we treat four spectral parameters $\boldsymbol{\theta} = (A_{\text{eff}}, T_0, \mu, y)$, where $A_{\text{eff}} = A_{\alpha} D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ is the cumulative comoving foam variance at the CMB distance (Section V B 1), so that $\sigma_{\phi}^2(\nu) = (2\pi\nu/c)^2 A_{\text{eff}}$. Note that α is not fitted directly; only the lumped amplitude A_{eff} enters the fit, and bounds on α are obtained *a posteriori* by inverting the Ng–Perlman relation at fixed $A_{\alpha} = 1$.

In the physically-realized regime (Section V B 2) the redistributed pedestal lies at $\sim 10^{42} \text{ Hz}$, eleven orders of magnitude above the FIRAS band, so the only in-band foam signature is the pure deficit

$$\frac{\partial I_{\nu}}{\partial A_{\text{eff}}} = - \left(\frac{2\pi\nu}{c} \right)^2 B_{\nu}(T_0), \quad (38)$$

a ν^2 -weighted (blue-tilted) Planck spectrum. The chemical-potential and Compton- y distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0) \frac{x e^x}{(e^x - 1)^2}, \quad (39)$$

$$\frac{\partial I_\nu}{\partial y} = B_\nu(T_0) \frac{x e^x}{e^x - 1} \left[\frac{x(e^x + 1)}{e^x - 1} - 4 \right], \quad (40)$$

with $x = h\nu/(k_B T_0)$, encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas).

We constrain A_{eff} by fitting these templates — together with the FIRAS Galactic-foreground template — to the public COBE/FIRAS monopole residual spectrum and its calibration covariance [21]. Raising T_0 brightens the whole Planck curve, whereas the foam deficit suppresses it with an extra ν^2 weighting; over the CMB-bright band (the peak through the onset of the Wien fall-off) the two differ only by a spectral tilt, so the foam template is nearly indistinguishable from a temperature shift ($\rho_{A_{\text{eff}} T_0} \simeq 1$ in the fit) and, through that near-degeneracy, strongly collinear with the y and μ templates (quantified by the forecast correlation matrix Eq. (42) below). Using the full 43×43 apodization covariance, the generalized least-squares fit over $(T_0, \mu, y, A_{\text{eff}})$ and the foreground amplitude recovers μ and y consistent with zero and with the Fixsen limits ($|\mu| < 1.7 \times 10^{-4}$, $|y| < 1.0 \times 10^{-5}$ at 95%; $\chi^2/\nu = 1.24$) and a foam amplitude consistent with zero,

$$A_{\text{eff}} = (-2.4 \pm 5.3) \times 10^{-11} \text{ m}^2. \quad (41)$$

Inverting $A_{\text{eff}} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ at $A_\alpha = 1$, the one-sided 95% upper limit $A_{\text{eff}} < 1.645 \sigma(A_{\text{eff}}) = 8.7 \times 10^{-11} \text{ m}^2$ gives the lower bound $\alpha \gtrsim 0.52$ (collected with the forecasts below in Table I). The one direction that could still break the foam- T_0 degeneracy is deep into the Wien/sub-mm regime, where the ν^2 -tilt finally separates from a temperature shift, but galactic dust and the cosmic infrared background dominate — beyond the FIRAS band, and their foreground confusion makes the monopole impractical for breaking the degeneracy there. The linearised template $\Delta I_\nu \approx \sigma_\phi^2(\nu)$ [template], valid for $\sigma_\phi^2 \ll 1$, holds across the FIRAS band at the headline threshold (σ_ϕ^2 exceeds unity only for stronger foam, $\alpha \lesssim 0.50$; Section V B 1): replacing the deficit $-\sigma_\phi^2 B_\nu$ by the saturated form $-(1 - e^{-\sigma_\phi^2}) B_\nu$ changes the integrated in-band amplitude by $\approx 13\%$ at $\alpha = 0.50$ but by $< 1\%$ at $\alpha \gtrsim 0.52$ ($\Delta\alpha \sim 3 \times 10^{-5}$), so the bound is unaffected; nonlinearity matters only for stronger foam ($\alpha \lesssim 0.50$), where it may also help break the foam- μ degeneracy.

That same near-degeneracy has two consequences for the CMB temperature itself. First, admitting foam as a free parameter inflates the marginalized temperature uncertainty: the FIRAS spectral-shape determination of T_0 degrades from $1.2 \times 10^{-5} \text{ K}$ to $7.7 \times 10^{-4} \text{ K}$ (a factor ~ 64 ; $\sigma(y)$ and $\sigma(\mu)$ inflate by ~ 17 and ~ 3), the latter comparable to the full COBE/FIRAS calibration uncertainty $5.7 \times 10^{-4} \text{ K}$ [22] — so a foam-agnostic monopole analysis roughly doubles the temperature error budget, and the quoted precision of $T_0 = 2.72548 \text{ K}$ implicitly assumes no foam. Second, foam present but *omitted* from the fit shifts the central value by $\Delta T_0 = b_{T_0} A_{\text{eff}}$ with $b_{T_0} \approx -1.5 \times 10^7 \text{ K m}^{-2}$; for $A_\alpha = 1$ this is only $\approx 3 \times 10^{-22} \text{ K}$ at the holographic $\alpha = 2/3$ and $\approx 4 \times 10^{-63} \text{ K}$ at $\alpha = 1$, reaching $\sigma(T_0)$ only at $\alpha \approx 0.52$ where the foam is already a many- σ signal. Both effects therefore matter only for foam near FIRAS detectability; the recovered T_0 is immune to any foam consistent with the GRB bound $\alpha \gtrsim 0.63$.

4. Fisher forecasts for future spectrometers

To forecast the reach of future absolute spectrometers we propagate the same no-pedestal templates through a Fisher matrix. Calibrated to FIRAS with a flat per-channel noise $\sigma_\nu = 0.01 \text{ MJy/sr}$ (a band-averaged stand-in for the measured errors), the Fisher forecast gives $\sigma(A_{\text{eff}}) = 2.2 \times 10^{-11} \text{ m}^2$ and the same $\alpha \gtrsim 0.52$, reproducing the data fit; the conditional value $\sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 8.2 \times 10^{-13} \text{ m}^2$ shows that marginalizing over the distortions costs a factor ~ 27 through the foam- T_0 collinearity. The fiducial degeneracy structure is captured by the correlation matrix for $(A_{\text{eff}}, T_0, \mu, y)$,

$$\rho_{ij}^{\text{fid}} = \begin{pmatrix} 1.000 & \mathbf{0.9997} & 0.928 & 0.999 \\ \mathbf{0.9997} & 1.000 & 0.934 & 0.998 \\ 0.928 & 0.934 & 1.000 & 0.915 \\ 0.999 & 0.998 & 0.915 & 1.000 \end{pmatrix}, \quad (42)$$

in which the foam deficit is almost perfectly aligned with a temperature shift ($\rho_{A_{\text{eff}} T_0} = 0.9997$, bold) and strongly collinear with the y and μ distortions (0.999 and 0.928). Defining the marginal detection SNR as $A_{\text{eff}}/\sigma(A_{\text{eff}})$, so that $\text{SNR} = 1.645$ marks the one-sided 95% CL threshold, Figure 1 plots it versus α , and Table I collects the thresholds.

TABLE I. Marginal foam-detection thresholds (one-sided 95% CL): *lower* bounds on the exponent α ($A_\alpha = 1$), since the foam signal $\sigma_\ell^2 \propto (\ell_P/D_C)^{2\alpha}$ decreases with α and a non-detection therefore excludes small α . FIRAS is from the data fit (Section V B 3); PIXIE, SuperPIXIE, and Voyage 2050 from the Fisher forecast scaled by their published μ -sensitivities [23]. The three forecast entries (0.545, 0.550, 0.554) round to 0.55; the thresholds increase monotonically.

	lower bound
FIRAS	$\alpha \gtrsim 0.52$
PIXIE	$\alpha \gtrsim 0.55$
SuperPIXIE	$\alpha \gtrsim 0.55$
Voyage 2050	$\alpha \gtrsim 0.55$

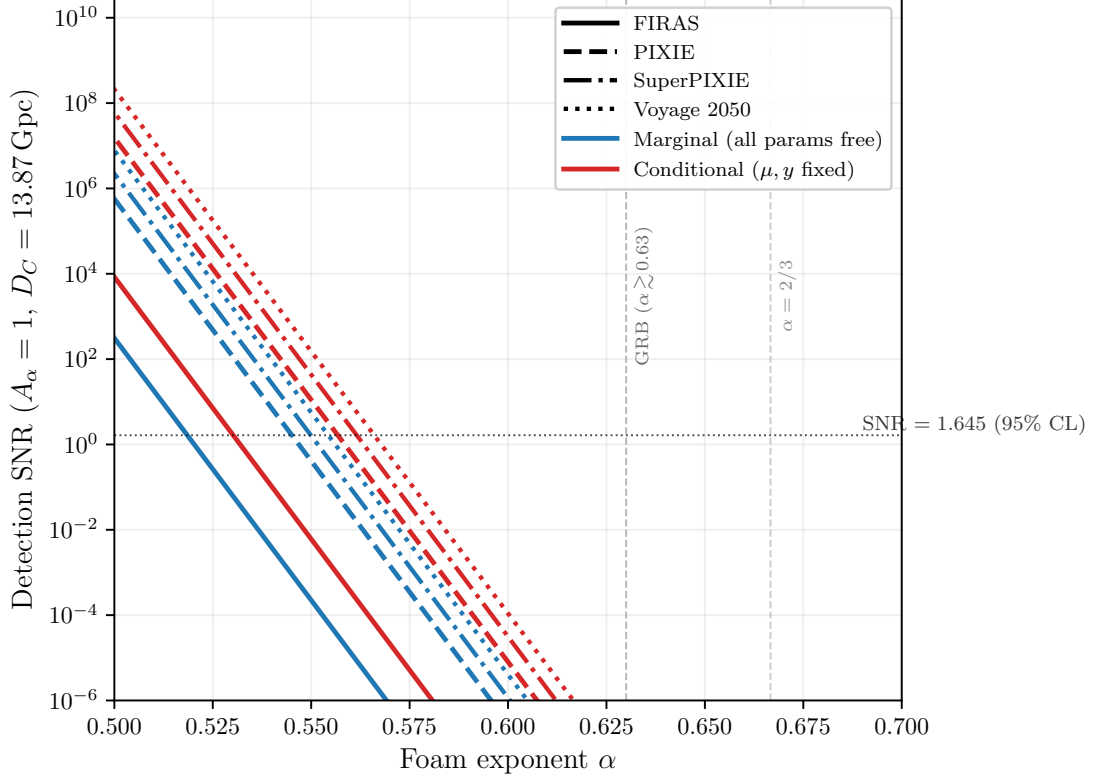


FIG. 1. Detection SNR for the foam amplitude A_{eff} as a function of foam exponent α ($A_\alpha = 1$, $D_C = 13.87$ Gpc). Blue curves: marginal SNR (all four parameters $A_{\text{eff}}, T_0, \mu, y$ free). Red curves: conditional SNR (μ, y held fixed), which is $\sim 27\times$ higher. Line styles denote the instrument: solid FIRAS, dashed PIXIE, dash-dotted SuperPIXIE, dotted Voyage 2050, with forecast μ -sensitivities scaled relative to FIRAS [23]. The marginal lower bounds (SNR = 1.645, one-sided 95% CL) cluster at $\alpha \approx 0.52$ – 0.55 , illustrating the logarithmic saturation of the bound with sensitivity. The dotted horizontal line marks SNR = 1.645 (one-sided 95% CL); vertical dashed lines mark the spectroscopic lower bound from GRB 221009A ($\alpha \gtrsim 0.63$) and the holographic prediction $\alpha = 2/3$.

To bound the impact of μ -aligned in-band pedestal contamination, we repeat the analysis with the μ -aligned band-filling kernel of Section V B 2, replacing the deficit by

$$\frac{\partial I_\nu}{\partial A_{\text{eff}}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0) + K_\nu^{\text{fast}}(\nu) \int_{\nu_{\text{min}}}^{\nu_{\text{max}}} \left(\frac{2\pi\nu'}{c}\right)^2 B_{\nu'}(T_0) d\nu', \quad (43)$$

whose $1/\nu$ pedestal we deliberately gave the Rayleigh–Jeans shape of the μ -distortion template (Eq. (36); the Rayleigh–Jeans limit of Eq. (39) is $\Delta I/B \rightarrow -\mu/x \propto \nu^{-1}$) so as to maximize the foam– μ overlap. The resulting correlation

matrix for $(A_{\text{eff}}, T_0, \mu, y)$ is

$$\rho_{ij}^{\mu\text{-align}} = \begin{pmatrix} 1.000 & 0.937 & \mathbf{0.976} & 0.010 \\ 0.937 & 1.000 & 0.969 & -0.212 \\ \mathbf{0.976} & 0.969 & 1.000 & -0.127 \\ 0.010 & -0.212 & -0.127 & 1.000 \end{pmatrix}. \quad (44)$$

Compared with the fiducial matrix Eq. (42), adding the in-band pedestal rotates the foam template off the temperature axis ($\rho_{A_{\text{eff}}T_0} : 0.9997 \rightarrow 0.937$) and off the y template ($\rho_{A_{\text{eff}}y} : 0.999 \rightarrow 0.010$), leaving the engineered foam- μ degeneracy ($\rho_{A_{\text{eff}}\mu} = 0.976$, bold) as the dominant one. Because 0.976 is milder than the 0.9997 it replaces, the amplitude is in fact *better* constrained,

$$\sigma(A_{\text{eff}}) = 1.7 \times 10^{-12} \text{ m}^2, \quad \sigma(A_{\text{eff}})|_{\mu, y \text{ fixed}} = 2.9 \times 10^{-13} \text{ m}^2, \quad (45)$$

a factor ~ 6 marginalization penalty driven by the foam- μ collinearity, and the threshold *tightens* slightly to $\alpha \gtrsim 0.53$. The μ -aligned in-band pedestal therefore does not loosen the bound: the factor $N \approx 13$ between the two amplitude uncertainties is compressed by the logarithmic Ng–Perlman relation $\sigma_\ell^2 = D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ into a negligible shift $\Delta\alpha = \ln N / [2 \ln(D_C/\ell_P)] \approx 0.009$. An external constraint on μ helps most in this μ -limited regime: imposing the Λ CDM dissipation prediction $\mu \approx 2 \times 10^{-8}$ [23] as a prior (five orders of magnitude tighter than the FIRAS μ measurement, hence effectively fixing μ) sharpens $\sigma(A_{\text{eff}})$ by ~ 3 – $5\times$, though the logarithmic relation again limits the threshold gain to $\Delta\alpha \lesssim 0.005$; in the physical no-pedestal case, where the limiting degeneracy is with T_0 rather than μ , even fixing μ barely helps. Either way, the worked example of Section VB 2 shows that any exponential-class C_α with Planck-scale τ_{foam} lies in the no-pedestal regime, so the physical $\alpha \gtrsim 0.52$ is the headline and the μ -aligned $\alpha \gtrsim 0.53$ merely confirms its robustness.

The proposed PIXIE satellite [24] — a fully cryogenic ($T_{\text{sys}} \approx 2.7$ K) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed Λ CDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived above, the marginal foam detection threshold shifts from $\alpha \gtrsim 0.52$ (FIRAS) to approximately $\alpha \gtrsim 0.55$ (PIXIE; Table I), narrowing the gap with the spectroscopic lower bound $\alpha \gtrsim 0.63$ from GRB 221009A (Fig. 1). All thresholds are quoted for the physical no-pedestal deficit. The step sizes track the logarithm of the sensitivity ratio — each decade of $\sigma(\mu)$ buys only $\Delta\alpha = \ln 10 / [2 \ln(D_C/\ell_P)] \approx 0.008$ — so the ~ 3 -decade FIRAS→PIXIE gain accounts for the ≈ 0.03 step, whereas PIXIE→Voyage 2050 (under a decade combined) adds only ≈ 0.009 ; the thresholds saturate monotonically (the three forecast entries round 0.545, 0.550, 0.554 to 0.55). The limiting degeneracy — with a temperature shift for the no-pedestal deficit — persists on the linearized template, but PIXIE’s extension to 6 THz accesses the nonlinear regime $\sigma_\phi^2 \gg 1$ for $\alpha \lesssim 0.52$ — at higher α than the FIRAS-band crossing $\alpha \lesssim 0.50$ (Section VB 3), since the wider 6 THz band pushes the $\sigma_\phi^2 = 1$ boundary upward — where the foam deficit diverges markedly from both the blackbody and μ -distortion shapes and the degeneracy may be substantially reduced. The $\alpha \gtrsim 0.52$ projection may therefore understate the reach: if the nonlinear template effectively breaks this degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent.

Extending the same Fisher analysis to the 10 GHz–3 THz bands of the SuperPIXIE and Voyage 2050 concepts [23] — with forecast distortion sensitivities $\sigma(\mu) \approx 7.7 \times 10^{-9}$ and 2.2×10^{-9} — shifts the marginal threshold only to $\alpha \gtrsim 0.55$ (Table I), barely beyond the PIXIE projection. The reason is structural rather than instrumental: in the physically realized no-pedestal limit the foam deficit $-\sigma_\phi^2(\nu)B_\nu$ is degenerate with a temperature shift at $\rho_{A_{\text{eff}}T_0}^{\text{fid}} = 0.9997$, and this degeneracy is *band-robust* — broadening the coverage from the FIRAS band to 10 GHz–6 THz leaves it unchanged to four decimal places. The wider band does break the foam- μ degeneracy probed by the μ -aligned K_ν^{fast} pedestal (from $\rho_{A_{\text{eff}}\mu}^{\mu\text{-align}} = 0.98$ at FIRAS to 0.66 over 10 GHz–3 THz), so future missions are markedly more robust against in-band pedestal contamination, but the physical threshold does not move. Together with the logarithmic Ng–Perlman scaling $\Delta\alpha = \ln N / [2 \ln(D_C/\ell_P)]$, which converts even a 10^4 -fold sensitivity gain over FIRAS into $\Delta\alpha \lesssim 0.04$, this sets a firm ceiling: as an independent broadband probe — absolute calibrated photometry, no source-profile assumption, and a different systematic-error budget from spectroscopy — spectral-distortion photometry reaches $\alpha \gtrsim 0.55$ and cannot, by sensitivity or bandwidth alone, attain the GRB spectroscopic bound $\alpha \gtrsim 0.63$ or the holographic value $\alpha = 2/3$. The two channels are therefore permanently complementary.

VI. CONCLUSIONS

Spacetime foam imprints an achromatic, stochastic arrival-time jitter that modifies the first-order temporal coherence $g^{(1)}(\tau)$, and hence the observed spectrum, providing a spectroscopic probe complementary to the imaging (PSF,

Strehl) channel. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a shape-distortion bound on the Ng–Perlman exponent α . The MeV emission lines of GRB 221009A give $\alpha \gtrsim 0.63$ (one-sided 95% CL), while a fit to the COBE/FIRAS monopole gives $\alpha \gtrsim 0.52$ (95% CL); next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) reach only $\alpha \gtrsim 0.55$, the bound saturating logarithmically because the foam deficit is nearly degenerate with a temperature shift. None of these bounds exclude the holographic value $\alpha = 2/3$.

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit $\tau_{\text{foam}} \ll \tau_c$ (Section III B), in which the foam path-length jitter decorrelates many times within each photon coherence time. If foam fluctuations are instead slow ($\tau_{\text{foam}} \gg \tau_c$), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation, and the shape-distortion bounds do not apply. The CMB constraint (Section V B) is similarly derived in the dynamical limit (Section V B 2 works in $C_\alpha \rightarrow 0$): in the opposite quasi-static limit the per-mode foam suppression $\mathcal{F}_{\text{foam}}(\tau) \rightarrow 1$ (Eq. (13)) and the CMB monopole is unmodified, so the shape-distortion bounds do not apply in either channel.

The dynamical bound $\alpha \gtrsim 0.63$ (Eq. (28)) does not exclude the holographic value $\alpha = 2/3 \approx 0.667$. A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ($\lesssim 40$ MeV, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ($\gtrsim 100$ MeV). The frequency $\nu_\star \simeq 37$ MeV at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from ~ 100 keV to beyond 1 GeV. The All-sky Medium Energy Gamma-ray Observatory (AMEGO [25]) and its MIDEX-scale variant AMEGO-X [26] are designed to cover 100 keV to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [27] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch ~ 2027) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder, though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work; reducing the per-epoch line-width uncertainties to $\delta r \lesssim 0.01$ would tighten the bound to $\alpha \gtrsim 0.64$, removing a major systematic and enabling substantially tighter tests of spacetime-foam models. Independently, a joint fit of the line centroid and width across epochs — the present analysis constrains the width alone — would exploit the energy-dependent line-center shift $\nu_0^{\text{eff}} = \nu_0 \tau_G^2 / (\tau_G^2 + \sigma_\Delta^2)$ (Eq. (21)) that the same foam jitter induces, adding a second handle on α and further tightening the bound.

A complementary implication runs in the opposite direction. The flagship goal of next-generation CMB spectrometers — PIXIE, SuperPIXIE, Voyage 2050 — is a first detection of the chemical potential μ , whose guaranteed Λ CDM value $\mu \simeq 2 \times 10^{-8}$ arises from Silk damping of primordial perturbations. Our Fisher analysis shows that the foam-induced distortion is nearly degenerate with the μ template across the accessible band (Section V B 3); the two cannot be separated from the monopole spectrum alone. A measured μ that *departs* from the Λ CDM prediction would be a landmark signal whatever its origin, and spacetime foam is one propagation-induced alternative to early-universe energy injection — distinguishable only with external information. The GRB lower bound $\alpha \gtrsim 0.63$ already limits any foam contribution to the CMB monopole to far below the Λ CDM μ , so foam does not contaminate the guaranteed- μ target; but this same spectroscopy-plus-GRB complementarity is what would be needed to attribute a larger, anomalous μ . In this sense the CMB μ -distortion measurement and the foam search are two readings of a single spectral degeneracy. The decisive limitation here is theoretical rather than instrumental: the pedestal shape — and hence how completely the foam deficit can be separated from a temperature shift or from μ — is fixed by the foam temporal autocorrelation $C_\alpha(\tau)$, which still awaits a first-principles derivation from a quantum-gravity model.

The multi-epoch r_{obs} dataset constrains any propagation effect that broadens an emission line in proportion to a power of E , not only the Ng–Perlman spacetime-foam template considered here. The most natural extension is to stochastic Lorentz-invariance violation [16], in which the photon group velocity carries a Planck-suppressed random coefficient $v(E) = c[1 - \xi(E/E_{\text{QG},n})^n]$, with ξ of order unity drawn independently for each Planck-scale traversal. Photons in the same line then propagate at slightly different speeds, broadening the observed width as $\sigma_{\text{LIV}}(E)/E \propto E^n$ on top of any intrinsic contribution. Unlike the deterministic LIV signal probed by photon time-of-flight studies — where GRB 221009A LHAASO TeV photons already constrain $E_{\text{QG},1} \gtrsim \text{a few} \times E_{\text{Pl}}$ [28] — stochastic LIV produces a *width* signal that survives light-curve cross-correlation, and a multi-epoch r_{obs} measurement is therefore one of the few clean probes available. Fitting Eq. (27) recovers the foam constraint as the $n = 1$ case; refitting the same seven-epoch dataset under $r_{\text{LIV}} \propto (E/E_0)^n$ would yield bounds on $\langle \xi^2 \rangle^{1/2} / E_{\text{QG},n}^n$ at each order, with the $n = 2$ (quadratic) case particularly valuable because the deterministic time-of-flight tests are weakest there

at $z = 0.151$.

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